

MANTA

One continuous ribbon. One gesture.

Cut like a baguette so it can be printed at home and still hold an adult.



■ IDEA

MANTA is a single thick ribbon: floor, dished seat, cantilevered backrest, forward-curved crown, tail-brace, back to the floor — no separate legs, no frame, no visible assembly. The name is literal: forked feet read as fins, the tail-brace as a fluke. The ocean makes most of the oxygen we breathe, and the manta ray is one of its health indicators — so the surface carries the story too: flutes that follow the ribbon's own motion, bold at the floor and easing toward the crown, capped near the seat and back so they stay tactile without ever pressing into a bone.

■ BIOMIMICRY AS STRUCTURE, NOT ORNAMENT

Every centimetre of the section is set by a biological rule, and each is load-bearing:

- **Wolff's law** — thickness follows the bending moment: 90 mm at the lumbar knot, 66 mm at the knee, a 26 mm blade at the crown.
- **Murray's law** — the lumbar, the one true junction, is the cube-root-sum of seat, backrest and tail; every transition filleted.
- **Spiral fibre** — the flutes follow the stress path and stiffen by corrugation; the section twists up to 20° through the backrest.
- **Shell curvature** — the closed superelliptical section and dished seat carry load in membrane action.
- **Fiddlehead scroll** — the forward-curved crown closes the backrest into a tube, not an open cantilever.
- **Buttress roots** — each foot forks into two splayed prongs.

■ PRINTING

41 segments, each within 212 × 220 × 250 mm, printable **without supports** (8 print upright) — every gram becomes chair, none is scrapped as support waste for an ocean that has enough plastic in it already. PETG, 0.2 mm layers, 5 perimeters, 35 % gyroid — kept low, thickness already carries the load. Finished weight ~16 kg, ~180 h batch print time, ~90 € in filament.

■ ASSEMBLY — SCREW-FREE

The ribbon is sliced **crosswise**, so every joint is a full solid cross-section (7,000–17,000 mm²), never a thin plate edge: a self-centring conical tenon (Ø32) plus epoxy. Wide segments also split along the centre-line, seam hidden in a flute, reading as a keel. The two joints beside the lumbar get an added Ø10 pin. No screws, no brackets.

■ STABILITY AND LOAD

First-order check (120 kg × 1.8 dynamic × 2.0 safety) re-derived live from the width/thickness tables: every member passes, weakest link 1.6× without glue (confirmed by hand-breaking a printed coupon that held on the pin alone), 4.8× at the weakest epoxied joint. Tipping 0.56 sideways, 0.55–0.77 rearward, safe static load ~190 kg. Seat 430 × 430 mm, 455 mm high, 3° rearward tilt, convex lumbar.

192 kg

safe static load

4.8×

weakest epoxied joint

~16 kg

finished weight

~90 €

filament cost

■ OPEN DESIGN — NOTHING REPEATS

Nothing in nature repeats, so no two parts of this chair do either: all 41 segments are geometrically unique (only the small locking pin repeats). One parametric script: change the tables and it regenerates a leaner variant with the same joints — automatically re-checked against the same load case, so it can't silently export a variant that fails. Full source, anonymised for review: anonymous.4open.science/r/manta-chair-23D4